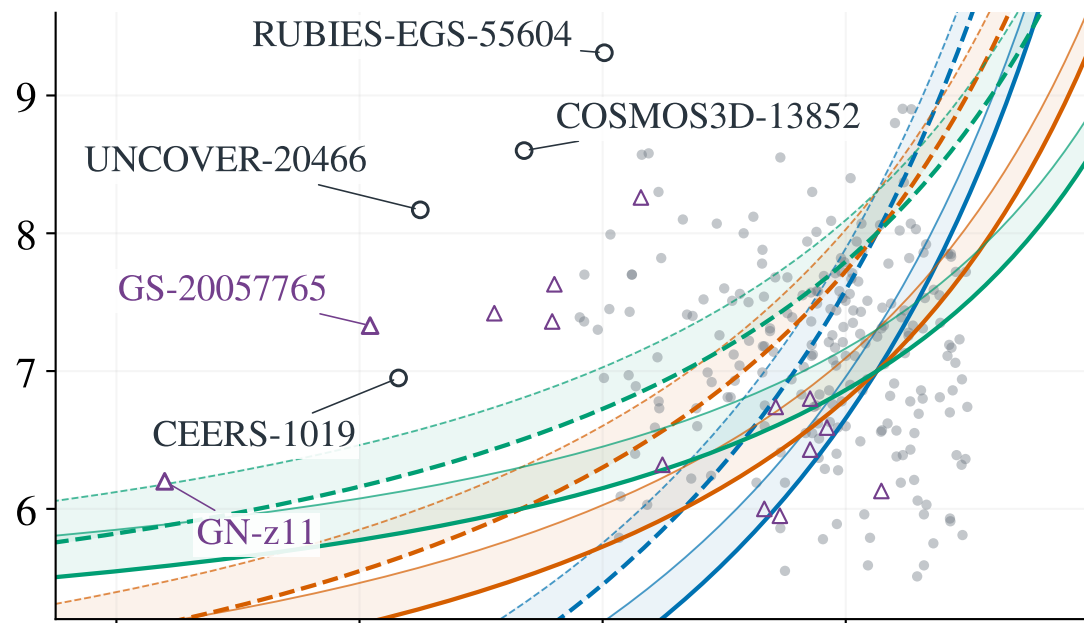


Growth tracks across radiative-efficiency assumptions

$z_{\text{seed}} = 30$; bands span $B_{\text{merge}} = 1$ (thick edge) to $B_{\text{merge}} = 2$ (thin edge).

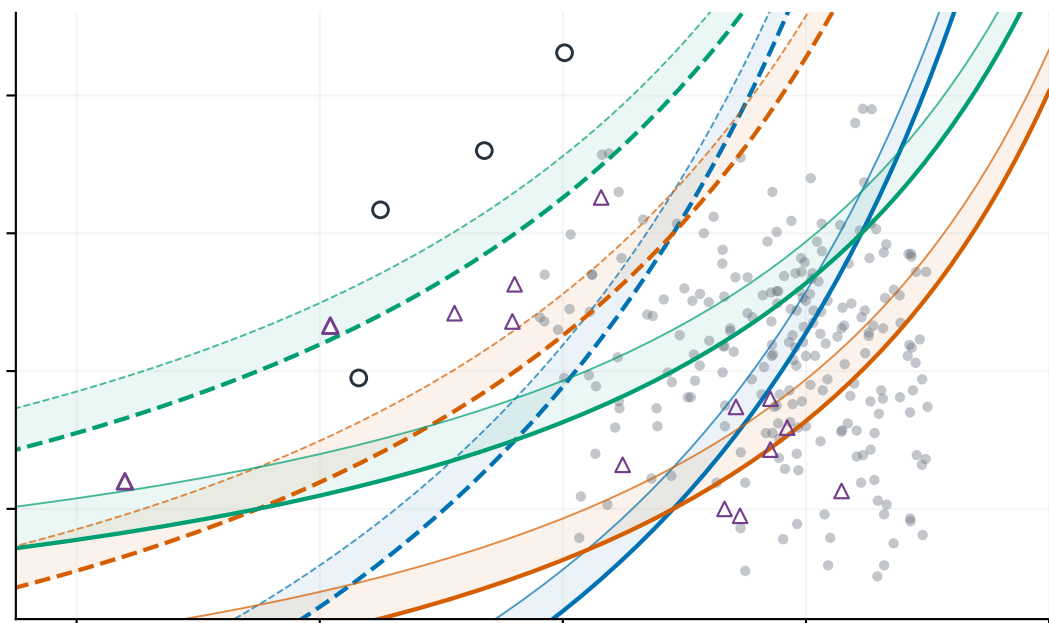
(a) $\varepsilon = 0.10000$
 $M_{\text{seed}} : \bar{f}_{\text{Edd}}$

— $10^2 M_{\odot} : 0.5$ — $10^4 M_{\odot} : 0.4$
- - $10^2 M_{\odot} : 0.6$ — $10^5 M_{\odot} : 0.2$
— $10^4 M_{\odot} : 0.3$ - - $10^5 M_{\odot} : 0.3$



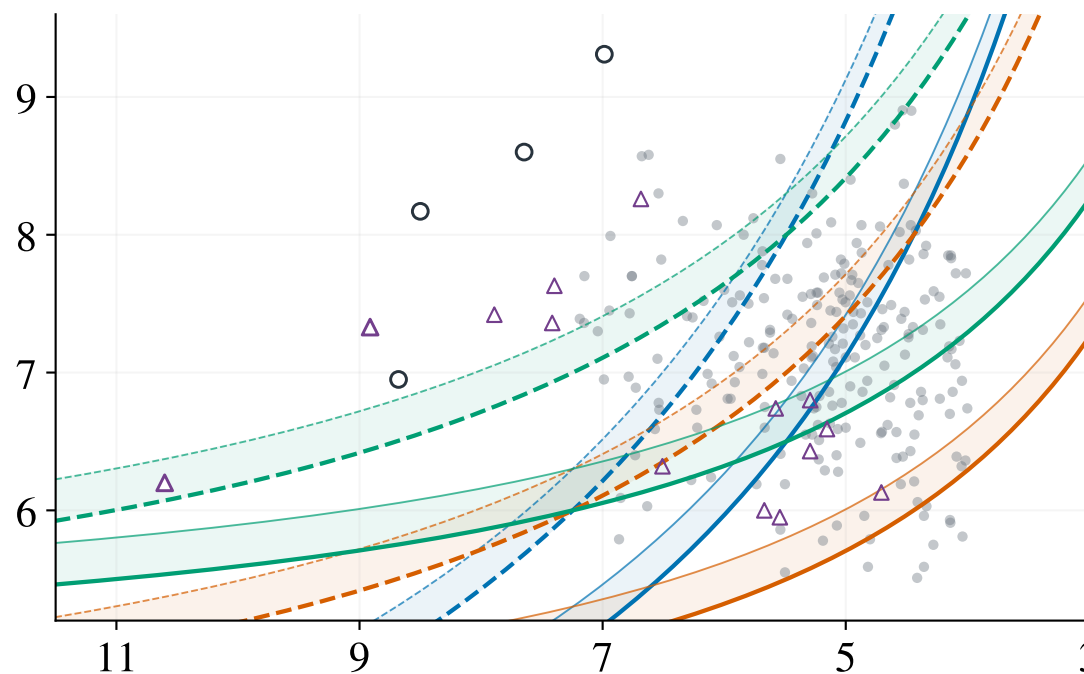
(b) $\varepsilon = 0.03775$
 $M_{\text{seed}} : \bar{f}_{\text{Edd}}$

— $10^2 M_{\odot} : 0.2$ — $10^4 M_{\odot} : 0.2$
- - $10^2 M_{\odot} : 0.3$ — $10^5 M_{\odot} : 0.1$
— $10^4 M_{\odot} : 0.1$ - - $10^5 M_{\odot} : 0.2$



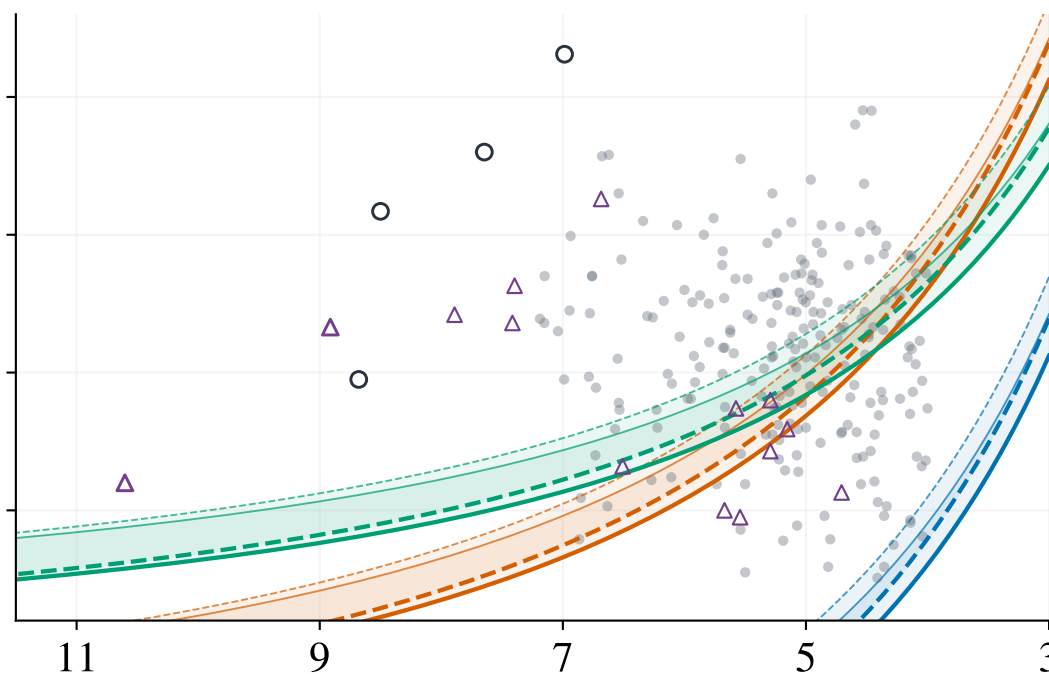
(c) $\varepsilon = 0.05719$
 $M_{\text{seed}} : \bar{f}_{\text{Edd}}$

— $10^2 M_{\odot} : 0.3$ — $10^4 M_{\odot} : 0.2$
- - $10^2 M_{\odot} : 0.4$ — $10^5 M_{\odot} : 0.1$
— $10^4 M_{\odot} : 0.1$ - - $10^5 M_{\odot} : 0.2$



(d) $\varepsilon = 0.42265$
 $M_{\text{seed}} : \bar{f}_{\text{Edd}}$

— $10^2 M_{\odot} : 1.9$ — $10^4 M_{\odot} : 2$
- - $10^2 M_{\odot} : 2$ — $10^5 M_{\odot} : 1.3$
— $10^4 M_{\odot} : 1.9$ - - $10^5 M_{\odot} : 1.4$



Observed redshift

Grey circles: primary (220); purple triangles: expanded sample only (14).

Outlined targets are named in panel (a).